



ISLAMIC UNIVERSITY OF TECHNOLOGY



Course Outline and Course Plan (MATH 4643)

Academic Session: 2023-2024

Name of the Teacher	Dr. Muntasir Alam	Position	Assistant Professor (Part-Time)
Department	Computer Science and Engineering	Program	B. Sc. in SWE
Course Code	MATH 4643	Course Title	Probability and Statistics II
Academic Year	2023-2024	Semester	4 th
Contact Hours	3.0	Credit Hours	3.0
Textbooks and Authors of the Reference books	1. Probability and Statistics for Engineers and Scientists, Academic Press, 4th Edition, 2009.; <i>Sheldon M. Ross</i>. 2. Probability and Stochastic Process, Wiley, 3rd Edition, 2014; <i>Roy D. Yates and David J. Goodman</i>. 3. Introduction to probability, Taylor and Francis; <i>Joseph K. Blitzstein and Jessica Hwang</i>.		
Prerequisites	N/A		
Curriculum Requirement	Compulsory		
Course Homepage			
Teaching Methods/ Approaches	☒ Lecture ☐ Project	☐ Group discussion ☐ Others: Presentation by Students	☒ Demonstration ☒ Problem solving
Teaching aids	☒ Multi-media ☐ OHP	☒ Board and Marker ☐ Others	

Course Assessment Method								
Attendance (10%)	Quiz 15% of Total Marks (Best 3 out of 4)						Mid Semester (25%)	Semester Final (50%)
Evaluate based on the participation in the class	1 st Quiz	2 nd Quiz	3 rd Quiz	4 th Quiz	Others		Week /Date	Week /Date
	Week /Date	Week /Date	Week /Date	Week /Date	Assignment	Homework		
	3 rd Week	6 th Week	10 th Week	13 th Week	Will be given accordingly	Will be given accordingly	As per the schedule of IUT	As per the schedule of IUT

Course Contents and Objectives	<u>Course Contents</u> Review of Probability: Independence, Total Probability Theorem, Bayes' Theorem, Probability Mass Functions (PMF), Cumulative Distribution Functions (CDF), Expectation, Variance, Joint PMFs, PDFs, Conditional Expectation; Random vectors; Stochastic process and their classifications, Bernoulli Process and Poisson Process, and their properties, Discrete-time and continuous-time Markov Process, Stationary probabilities and balance equation. Introduction to queuing theory, Hypothesis testing – Test concerning the mean and variance of the normal population, Regression, and correlation. Two-factor analysis of variance and two-way analysis of variance with interaction, Goodness of fit test with specified and unspecified parameters, Non-parametric hypothesis sing test. <u>Objectives</u> ➤ To learn the basic concepts addressed in advanced probability theory and statistical analysis. ➤ To learn how to deal with different types of stochastic processes in real-life problems using the knowledge of mathematics and computational statistics. ➤ To be able to understand the properties of queuing theory, hypothesis testing, regression, and correlation analysis. ➤ To familiarize students with the two-factor as well as two-way analysis of variance with interaction. ➤ To help students understand the importance of the goodness of fit test, a non-parametric hypothesis sign test appeared in the field of science and engineering.
Course Outcomes	<u>Course Outcomes (COs)</u> CO1 – Constructing the definition of the well-known statistical concepts along with necessary conditions or properties. CO2 – Applying the fundamental theories and models of probability and statistics in real-life applications especially encountered in the fields of science and engineering. CO3 – Representing and Solving real-life applications by using the fundamental tools of computational mathematics.

Weekly plan for course content		
Weeks	Topics	Task/Reading
1	Review of probability: Independence, Total Probability Theorem, Bayes' Theorem;	
2	Review of probability: Probability Mass Functions (PMF), Cumulative Distribution Functions (CDF), Expectation, Variance;	

3	Review of probability: Joint PMFs, PDFs, Conditional Expectation;	Quiz 1 – CO1, CO2, CO3.
4	Random vectors;	
5	Stochastic processes and their classifications;	
6	Bernoulli Process and Poisson Process, and their properties;	Quiz 2 – CO1, CO2, CO3.
7	Discrete-time and continuous-time Markov Process;	
	MID-Semester Examination	MID - CO1, CO2, CO3.
8-9	Stationary probabilities, and balance equation;	
10	Introduction to queuing theory;	Quiz 3 – CO1, CO2, CO3.
11	Hypothesis testing – Test concerning the mean and variance of the normal population;	
12	Regression, and correlation;	
13	Two-factor analysis of variance and two-way analysis of variance with interaction;	Quiz 4 – CO1, CO2, CO3.
14	Goodness of fit test with specified and unspecified parameters, Non-parametric hypothesis sing test;	
15	Review Class	
	Semester Final Examination	Final - CO1, CO2, CO3

Student Learning Time

NO.	Teaching and Learning Activities	Student Learning Time (SLT): Notional Hours
1	Lectures	45 hours
2	Tutorial	5 hours
4	Assignment/Problem solving/Tests	15 hours
5	Personal study	40 hours
6	Cooperative learning	15 hours
TOTAL SLT		120 hours
CREDIT = SLT/40		3

CO-PO Mapping with Bloom's Taxonomy

Mapping with CO, PO and Bloom's Taxonomy			
CO No.	Course Outcomes (CO) Statement	Levels of Bloom's Taxonomy	Matching with Program Outcome (PO)
CO1	Constructing the definition of the well-known statistical concepts along with necessary conditions or properties.	K1, C3	PO1
CO2	Applying the fundamental theories and models of probability and statistics in real-life applications especially encountered in the fields of science and engineering	K1, C3	PO2
CO3	Representing and Solving real-life applications by using the fundamental tools of computational mathematics.	K1, C3	PO3

Program Outcomes (POs: PO1 ~ PO12)

PO No.	Program Outcomes (POs)
	Students graduating from the Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE) program, upon graduation students will have the ability to:
PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences. (K1 to K4)
PO3	Design/development of solutions: Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)
PO4	Investigation: Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
PO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Grading Policy

Numeric Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Class Schedule

Section	Day	Time	Room Number
SWE	Friday (MUA)	08:00 – 09:15	304 (AB2)
SWE	Friday (MUA)	11:45 – 01:00	304 (AB2)

Student's Consulting Hour: Thursday at 10:00 AM - 10:30 AM & 12:15 PM - 01:00 PM

Instructor contact details:

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